Assignment-1: Solutions

Instructor: **Goutam Sheet**
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Solution 1

Note: In this problem, motion is "communicated" to the coin by means of friction. Let the card be snapped out to the right side. We take this direction as the positive x-axis, with the origin on the rim (left side) of the glass. In this case, the frictional force on the card acts to the left, and that on the coin acts to the right.

When the card is snapped out, the frictional force acting on the coin (to the right) is μmg, so the horizontal acceleration of the coin is

$$a = \mu g.$$

Let the coin be initially at position $x = r$ (distance r from the origin/rim). The coin's position at time t is

$$x(t) = r + \tfrac{1}{2}at^2 = r + \tfrac{1}{2}\mu g t^2.$$

If the card has (rear-edge) speed v then the time for the rear edge to cross the diameter $2r$ is

$$T = \frac{2r}{v}.$$

For the coin to fall into the glass its displacement at time T must place it inside the rim, i.e. $x(T) < 2r$. Compute

$$x(T) = r + \tfrac{1}{2}\mu g \left(\frac{2r}{v}\right)^2 = r + \tfrac{1}{2}\mu g \frac{4r^2}{v^2} = r + \frac{2\mu g r^2}{v^2}.$$

Require $x(T) < 2r$, so

$$r + \frac{2\mu g r^2}{v^2} < 2r \quad \Longrightarrow \quad \frac{2\mu g r^2}{v^2} < r \quad \Longrightarrow \quad v^2 > 2\mu g r.$$

Thus the minimum (snapping) velocity of the card is

$$\boxed{v_{\min} = \sqrt{2\mu g r}.}$$

Solution 2

Equation of motion of a boat subject to a resistive force proportional to its velocity is

$$m\frac{dv}{dt} = -\mu v,$$

(a) Separate variables and integrate:

$$\frac{dv}{v} = -\frac{\mu}{m}\,dt \quad \Longrightarrow \quad \ln v = -\frac{\mu}{m}t + C.$$

Using $v(0) = v_0$ gives $C = \ln v_0$, hence

$$\boxed{v(t) = v_0 e^{-\mu t/m}.}$$

First question: Let time is t_1 when $v = v_0/2$:

$$\frac{v_0}{2} = v_0 e^{-\mu t_1/m} \quad \Longrightarrow \quad e^{-\mu t_1/m} = \tfrac{1}{2} \quad \Longrightarrow \quad t_1 = \frac{m}{\mu}\ln 2.$$

Second question: Distance traveled during t_1:

Method 1

Equation of motion: $\qquad mv\dfrac{dv}{dx} = -\mu v,$

Assume $v \neq 0$ so we may divide by v : $\qquad m\dfrac{dv}{dx} = -\mu,$

Integrate with the initial condition $v(0) = v_0$:

$$\begin{aligned} m\int_{v_0}^{v} dv' &= -\mu\int_0^x dx' \\ m\big(v - v_0\big) &= -\mu x, \end{aligned}$$

hence $\qquad \boxed{v(x) = v_0 - \dfrac{\mu}{m}\,x\,.}$

In particular, setting $v(x_1) = \frac{v_0}{2}$ gives

$$\frac{v_0}{2} = v_0 - \frac{\mu}{m}x_1 \quad \Longrightarrow \quad \boxed{x_1 = \frac{mv_0}{2\mu}\,.}$$

Method 2

$$x_1 = \int_0^{t_1} v(t)\,dt = v_0\int_0^{t_1} e^{-\mu t/m}\,dt = v_0\left[-\frac{m}{\mu}e^{-\mu t/m}\right]_0^{t_1} = v_0\frac{m}{\mu}\left(1 - \frac{1}{2}\right) = \frac{v_0 m}{2\mu}.$$

Third question: Total distance before the boat "stops" (as $t \to \infty$):

$$x_2 = \int_0^{\infty} v_0 e^{-\mu t/m}\,dt = v_0\frac{m}{\mu}.$$

Note: The speed approaches zero as $t \to \infty$ (never reaches exact zero in finite time), but the total distance is finite.

Also note that the work done by friction equals the loss in kinetic energy:

$$W_{\text{fric}} = \tfrac{1}{2}mv_0^2 - \lim_{t\to\infty}\tfrac{1}{2}mv(t)^2 = \tfrac{1}{2}mv_0^2.$$

Solution 3

Equation of motion:

$$m\frac{dv}{dt} = -\beta v^2, \qquad v(0) = v_0.$$

Separate variables:

$$\frac{dv}{v^2} = -\frac{\beta}{m}\,dt.$$

Integrate from v_0 at $t = 0$ to v at time t:

$$-\frac{1}{v} + \frac{1}{v_0} = -\frac{\beta}{m}t \quad \Longrightarrow \quad \frac{1}{v} = \frac{1}{v_0} + \frac{\beta}{m}t$$

so

$$\boxed{v(t) = \frac{1}{\dfrac{1}{v_0} + \dfrac{\beta}{m}t} = \frac{mv_0}{\beta v_0 t + m}.}$$

From this equation it is evident that the speed continuously decreases with time and will be zero at $t = \infty$. In other words we can say that the particle will never come to rest.

Now for displacement $x(t)$,

$$\frac{dx}{dt} = v(t) = \frac{mv_0}{\beta v_0 t + m}.$$

Integrate: (with the assumption that $x = 0$ at $t = 0$)

$$x(t) = \int_0^t \frac{mv_0}{\beta v_0 t' + m}\,dt' = \frac{m}{\beta}\int_0^t \frac{\beta v_0}{\beta v_0 t' + m}\,dt' = \frac{m}{\beta}\left[\ln(\beta v_0 t' + m)\right]_0^t$$

$$\boxed{x(t) = \frac{m}{\beta}\ln\left(\frac{\beta v_0 t + m}{m}\right).}$$

Solution 4

With the point of projection as the origin, upward taken positive (x-axis) , the equation of motion during upward flight is modelled as

$$m\frac{dv}{dt} = -mg - mk^2v^2,$$

Write

$$\frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx},$$

so the differential equation becomes

$$v\frac{dv}{dx} = -\left(g + k^2v^2\right).$$

Separate variables:

$$\frac{v\,dv}{g + k^2v^2} = -dx.$$

Integrate from v_0 to 0 (at maximum height h) and $x : 0 \to h$:

$$\int_{v_0}^{0} \frac{v\,dv}{g + k^2v^2} = -\int_0^h dx = -h.$$

Let $u = g + k^2v^2$, $du = 2k^2v\,dv$, so

$$\int_{v_0}^{0} \frac{v\,dv}{g + k^2v^2} = \frac{1}{2k^2}\int_{g+k^2v_0^2}^{g} \frac{du}{u} = \frac{1}{2k^2}\ln\left(\frac{g}{g + k^2v_0^2}\right).$$

Thus

$$-h = \frac{1}{2k^2} \ln\left(\frac{g}{g + k^2 v_0^2}\right) \quad \Longrightarrow \quad \boxed{h = \frac{1}{2k^2} \ln\left(\frac{g + k^2 v_0^2}{g}\right).}$$

For the downward motion one uses (with the highest point as the origin, x-axis vertically downward)

$$v\frac{dv}{dx} = g - k^2 v^2,$$

and by integrating appropriately one obtains the speed v_1 when the projectile returns to the point of projection:

$$\boxed{v_1^2 = \frac{g v_0^2}{g + k^2 v_0^2}.}$$

(Note: in the limit $k \to 0$ (no air resistance) $v_1 \to v_0$, as expected.)

Solution 5

Consider an infinitesimal element of rope between x and $x + dx$. The weight of this segment is $\lambda g\, dx$ (or, if total mass is m and uniform, mass per unit length is m/l, so the weight is $(m/l) g\, dx$). Equilibrium of the element yields

$$T(x) = T(x + dx) + \frac{m}{l} g\, dx.$$

Rearrange and take limit $dx \to 0$:

$$\frac{dT}{dx} = -\frac{m}{l} g.$$

Integrate with boundary condition that the free end ($x = l$) is tension free: $T(l) = 0$. Integration gives

$$T(x) = -\frac{m}{l} g\, x + C.$$

Using $T(l) = 0$ gives $C = \frac{m}{l} g\, l = mg$. Thus

$$\boxed{T(x) = mg\left(1 - \frac{x}{l}\right) = \frac{mg(l - x)}{l}}$$

or, if you prefer linear density $\lambda = m/l$,

$$T(x) = \lambda g(l - x).$$

Remark: The method we have used here would have been essential if the rope had a variable density. The present case is simple and a simpler solution also possible.